# Honolulu – Tsunami ESF-6 playbook

## Readiness sheet

### Honolulu · Tsunami · Data as of 2026-10-28

**Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.** Per-area counts from 1 to 9 are not shown. All figures are public Census Bureau, CDC and HHS estimates, not personal records.

#### People in the risk area (county totals)

| Measure | People |
|---|---|
| People in the risk area | 80,344 ± 2,547 |
| May need a public shelter (low – high) | 1,607 – 8,838 |
| Speak English less than well | 3,374 ± 615 |
| Use power-dependent medical equipment (estimate) | 507 |
| People with a disability | 9,409 ± 712 |
| Age 65 and older, living alone | 3,841 ± 376 |
| People in households without a car | 10,265 ± 977 |

± approximate 90% margin of error of the county sum (Census Bureau method: square root of the summed squares of each Census area's margin, scaled to the risk area). None for power-dependent counts. Sources and years: section 8.

#### Usable 2026 sites for this hazard

George Fred Wright Wahiawā District Park, Kahuku Elementary School, Kāneʻohe District Park, Mānoa Valley District Park, Nānākuli Elementary School, Nānākuli High and Intermediate School.

#### Top 3 stress-test findings

Stress test, not a forecast.

1. **Extreme tsunami zone (worst case) – Honolulu:** 208,201 people in the extreme tsunami evacuation zone (may need a public shelter (likely): 10,410; 2026 sites inside the extreme zone: 1).
2. **Visitors on the island(s) – Honolulu (tsunami):** 107,479 visitors on an average day (island level) (visitors inside the tsunami zone: needs county data (hotel and rental locations)).
3. **Stranded visitors for 3 or 7 days – Honolulu:** 107,479 visitors on an average day (island level).

#### Partner checklist

| Role | Before | During | After |
|---|---|---|---|
| **County emergency management agency** | Confirm which sites open for this hazard and their capacity. | Decide which shelters open; announce them. | Decide when each shelter closes; ask for a plan rerun. |
| **Hawaiʻi DHS (State mass care)** | Share this sheet with partners; agree who leads each function. | Track open shelters and gaps; ask partners for help. | Coordinate moves to temporary housing and referrals to programs. |
| **Red Cross** | Check staff, cots and supplies against the likely demand. | Run shelters and feeding as agreed with the county. | Casework and referrals as agreed with the county. |
| **211 helpline** | Load shelter, transport and language information. | Answer calls; refer people to open shelters and transport. | Refer people to recovery programs and agencies. |
| **Faith and cultural groups** | Name trusted messengers for the languages in this playbook. | Share official messages; check on members who need help. | Help residents reach recovery services. |
| **Interpreters** | Confirm interpreters for the top languages and sign language. | Interpret at shelters and for official messages. | Interpret at recovery centers and for applications. |
| **Paratransit and accessible transport** | Confirm accessible vehicles and drivers. | Move people without a car or with mobility needs to shelters. | Return trips home or to temporary housing. |

Roles are proposals, to be confirmed with each partner. Details: sections 5 and 6.

**Data as of 2026-10-28**

**County:** Honolulu · **Hazard:** Tsunami

**Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.**

## 1. Summary

**Calculated by open code from public data · see Sources (section 8)**

| | | |
|---|---|---|
| People in the risk area<br>**80,344** | May need a public shelter (likely)<br>**4,017** | May need a public shelter (low – high)<br>**1,607 – 8,838** |
| Speak English less than well (C)<br>**3,374** | Use power-dependent medical equipment (estimate) (M)<br>**507** | People with a disability (I)<br>**9,409** |
| Age 65 and older (I)<br>**14,869** | People in households without a car (T)<br>**10,265** | Hearing difficulty (C)<br>**3,085** |
| Vision difficulty (C)<br>**1,591** | Cognitive difficulty (memory, decisions) (I)<br>**4,238** | Age 65 and older, living alone (S)<br>**3,841** |
| Known 2026 shelter sites usable for this hazard<br>**6** | People in the extreme tsunami zone (worst case)<br>**208,201** | |

Source: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER, FEMA National Risk Index v1.20; shelter rates: section 8.

Share of people in the risk area who may need a public shelter: low 2%, likely 5%, high 11%. One person can be counted in more than one need. The reading guide after section 2 explains these numbers.

## 2. Scope: county total vs. one community

**The county total is an upper bound.** It counts the people living inside the tsunami evacuation zone as hit at the same time. Real events usually hit fewer communities.

## Largest single community (the Census areas with the most people who may need a shelter)

| Community (Census area) | People in the risk area | May need a public shelter (low – likely – high) | Speak English less than well | Use power-dependent medical equipment (estimate) | People with a disability | Age 65 and older | People in households without a car |
|---|---|---|---|---|---|---|---|
| Māʻili<br>Census area 96.10 | 6,167 ± 1,333 | 123 – **308** – 678 | 16 ± 88 | 33 | 765 ± 333 | 960 ± 299 | 103 ± 166 |
| Lāʻie<br>Census area 102.02 | 5,026 ± 783 | 101 – **251** – 553 | 59 ± 60 | 14 | 328 ± 92 | 419 ± 113 | 508 ± 325 |
| Waiʻanae<br>Census area 97.07 | 3,933 ± 639 | 79 – **197** – 433 | 40 ± 53 | 18 | 700 ± 178 | 542 ± 150 | 1,109 ± 384 |

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

Per-area figures – People in the risk area: ACS 2018–2022, table S0601 (via CDC/ATSDR SVI 2022); Speak English less than well: ACS 2018–2022, table B16005 (via CDC/ATSDR SVI 2022); Use power-dependent medical equipment (estimate): HHS emPOWER county count, allocated by each area's share of people age 65 and older (estimate; no margin of error); People with a disability: ACS 2018–2022, table DP02 (DP02_0072E, via CDC/ATSDR SVI 2022); Age 65 and older: ACS 2018–2022, table S0101 (S0101_C01_030E, via CDC/ATSDR SVI 2022); People in households without a car: ACS 2018–2022, table DP04 (DP04_0058E, households without a vehicle, via CDC/ATSDR SVI 2022) × people per household; margin of error scaled the same way. ± is the 90% margin of error published with the estimate (SVI M_ columns; ACS), scaled by the fraction of residents who live in the risk area. Shelter estimates and need totals are model results with no margin of error. Per-area counts from 1 to 9 are not shown. All figures are public Census Bureau, CDC and HHS estimates, not personal records.

To plan for one community, use the first row.

# How to read this playbook

## Example row

| Community (Census area) | People in the risk area | May need a public shelter (low – likely – high) |
|---|---|---|
| Māʻili<br>Census area 96.10 | 6,167 ± 1,333 | 123 – **308** – 678 |

Read it as: about 6,167 people live in the risk area of Māʻili. If this hazard hits, between 123 and 678 of them may need a public shelter; 308 is the likely number. The ± after a number is its 90% margin of error.

## Census area

A neighborhood-sized area the U.S. Census uses, usually 1,200–8,000 people. The number (for example "Census area 308") is its Census ID. Each area is named after the Census place where most of its people live. "near" means no one in the area lives inside a named place; "rural near" means most people live outside named places.

## Low – likely – high

The share of people in the risk area who may go to a public shelter: low 2%, likely 5%, high 11%. Where the rates come from: Team assumptions: likely 5%; low 2% (the hurricane planning factor); high 11% (the Lahaina 2023 rate).

## Who has access and functional needs (CMIST)

- **C – Communication:** people who speak English less than well (they may need interpreters and translated messages), and people with hearing or vision difficulty.
- **M – Medical:** people who use power-dependent medical equipment (an estimate from HHS emPOWER).
- **I – Independence:** people with a disability, people with cognitive difficulty (memory, decisions) and people age 65 and older.
- **S – Safety and support:** people age 65 and older living alone, children and people living in group quarters (in the needs table).
- **T – Transportation:** people in households without a car.

## County total vs. one community

The county total counts the people living inside the tsunami evacuation zone as hit at the same time, so it is an upper bound. Use the "Largest single community" table to plan for an event that hits one community.

## Map

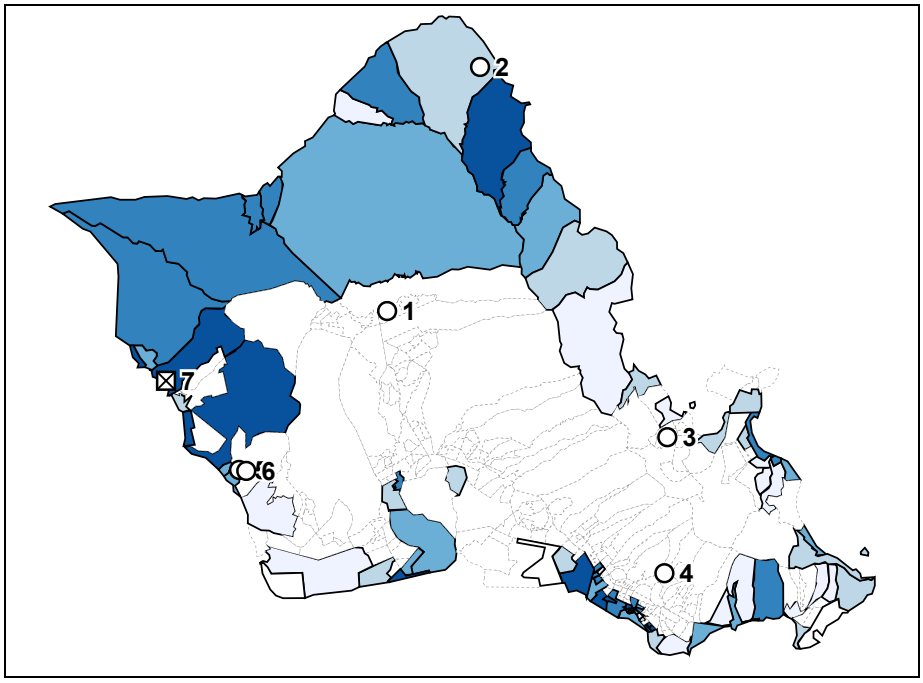
Darker blue means more people with needs, from "Lowest need" to "Highest need". A solid outline marks a Census area in the risk area; a dashed outline is outside it. Symbols: ○ usable site, △ check flood zone, ◇ flood risk not mapped, ⊠ not for this hazard. Numbers on the map match the site key.

## Why population numbers can differ

People in the risk area come from ACS 2018–2022 estimates (through the CDC Social Vulnerability Index 2022). Community names use 2020 Census counts. County totals in the plan use Census estimates for 2025. Each table names its source underneath.

**Every number is a planning estimate. In an event, follow county announcements.**

## 3. Map: people with needs and known sites




### Site key

**1** = George Fred Wright Wahiawā District Park (usable)

**2** = Kahuku Elementary School (usable)

**3** = Kāneʻohe District Park (usable)

**4** = Mānoa Valley District Park (usable)

**5** = Nānākuli Elementary School (usable)

**6** = Nānākuli High and Intermediate School (usable)

**7** = Waiʻanae District Park (not for this hazard)

- Lowest need (0 to 40 people)
- Lower (40 to 109 people)
- Middle (109 to 199 people)
- Higher (199 to 461 people)
- Highest need (461 to 1,868 people)
- Solid outline: Census area in the risk area
- Dashed outline: outside the risk area
- Usable site
- Check flood zone
- Flood risk not mapped (FEMA zone D)
- Not for this hazard

Color = people in the risk area who speak English less than well, use power-dependent medical equipment, have a disability, or live without a car (a person can be counted more than once). No street map: Census area outlines only.

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

**No 2026 activated site within a 15-minute drive (normal traffic)**, for sites usable for tsunami: 174 of 322 populated Census areas. With DOE candidate schools usable for this hazard: 17 of 322. This is not a statement that the area has no shelter: counties designate and open shelters for each event, and many public buildings are not on the 2026 list.

Drive-time matrix, per hazard: shortest drive from the area's land point to a 2026 activated site usable for that hazard on the same island; scenario 2 adds the 5 nearest DOE candidate schools usable for it. Site locations: U.S. Census Geocoder, checked against Esri ArcGIS World Geocoder. Travel times: Esri ArcGIS routing, normal traffic. Powered by Esri. Geocoding and routing: Esri ArcGIS Location Services.

## 4. Known sites (opened in 2026 events)

**Compiled from official county releases · verified by team**

| # | Status for this hazard | Site | Address | Why | Times opened in 2026 |
|---|---|---|---|---|---|
| 1 | ✓ **Usable** | George Fred Wright Wahiawā District Park | 1139-A Kilani Avenue, Wahiawā 96786 | Outside the tsunami evacuation zone | 1 |
| 2 | ✓ **Usable** | Kahuku Elementary School | | Outside the tsunami evacuation zone | 1 |
| 3 | ✓ **Usable** | Kāneʻohe District Park | 45-660 Keaʻahala Road, Kāneʻohe 96744 | Outside the tsunami evacuation zone | 1 |
| 4 | ✓ **Usable** | Mānoa Valley District Park | 2721 Kaʻaipū Avenue, Honolulu 96822 | Outside the tsunami evacuation zone | 1 |
| 5 | ✓ **Usable** | Nānākuli Elementary School | 89-778 Haleakala Avenue, Waiʻanae | Outside the tsunami evacuation zone | 1 |
| 6 | ✓ **Usable** | Nānākuli High and Intermediate School | | Outside the tsunami evacuation zone | 1 |
| 7 | ✕ **Not for this hazard** | Waiʻanae District Park | 85-601 Farrington Highway, Waiʻanae 96792 | Inside the tsunami evacuation zone | 1 |

Source: county announcements of sites opened in 2026; FEMA flood zones (Hawaiʻi Statewide GIS, State DFIRM); tsunami evacuation zones (Hawaiʻi Statewide GIS).

Site locations: U.S. Census Geocoder, checked against Esri ArcGIS World Geocoder. Travel times: Esri ArcGIS routing, normal traffic. Powered by Esri. Geocoding and routing: Esri ArcGIS Location Services.

Candidate public schools in this county usable for this hazard: **135**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

## Care facilities in hazard zones

**Care facilities in hazard zones – Honolulu, Tsunami (34 of 327), all in: Tsunami evacuation zone**

- Liberty Dialysis Hawaii - Hawaii Kai Dialysis (Dialysis facility)
- Liberty Dialysis Hawaii - Waianae Dialysis Facility (Dialysis facility)
- Liberty Waianae Coast Nanakuli (Dialysis facility)
- USRC Waianae Dialysis (Dialysis facility)
- Hale Malamalama (Nursing home)
- A Caring Place Montessori Preschool (Childcare centre)
- Ala Wai Elementary Head Start (Childcare centre)
- First Baptist Preschool Of Haleiwa (Childcare centre)
- HCAP Head Start 442 (Childcare centre)
- HCAP Head Start Hauula Elementary School (Childcare centre)
- HCAP Head Start Iroquois Point (Childcare centre)
- HCAP Head Start Laie Elementary (Childcare centre)
- HCAP Head Start Nanakuli I & II (Childcare centre)
- Haleʻiwa Elementary School (Childcare centre)
- Holy Nativity School (Childcare centre)
- Jefferson II Head Start (Childcare centre)
- Jefferson Pre-Plus Head Start (Childcare centre)
- Kamaaina Kids - Maili Preschool (Childcare centre)
- Kamaaina Kids Holy Trinity Preschool (Childcare centre)
- Kamehameha Preschool - Haleiwa I & II (Childcare centre)
- Kamehameha Preschool - Hoaliku Drake (Childcare centre)
- Kamehameha Preschool - Nanakuli II (Childcare centre)
- Kamehameha Preschool Nanakuli (Childcare centre)
- Kilohana United Methodist Church Preschool Inc. (Childcare centre)
- Metro Christian Academy (Childcare centre)
- PACT Kahauiki Village Head Start (Childcare centre)
- Poppins Keiki Hawaii (Childcare centre)
- Rainbow School Honolulu (Childcare centre)
- Seagull School Aina Haina (Childcare centre)
- St. Anthony School Early Learning Center (Childcare centre)
- Stepping Stones Academy Kakaako (Childcare centre)
- The Cole Academy Queen's Court (Childcare centre)
- Waialua Elementary School (Childcare centre)
- Waikiki Community Center Preschool - GCC (Childcare centre)

Source: CMS Provider Data Catalog (dialysis facilities, nursing homes, hospitals; facility addresses only), located with Esri ArcGIS World Geocoder; zones: State tsunami evacuation zone and FEMA flood zones (Hawaiʻi Statewide GIS, State DFIRM).

Source: Ready Keiki Locator, all pre-K sites (Office of the Lieutenant Governor, published through Hawaiʻi Statewide GIS): licensed group child care, public pre-K and charter pre-K sites only – no family child care homes; zones: State tsunami evacuation zone and FEMA flood zones (Hawaiʻi Statewide GIS, State DFIRM).

## 5. Needs and proposed partners

Likely estimate. Roles are **proposed – to be confirmed with partner**. The pledged column stays blank until a partner confirms it.

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Cots / sleeping space | 4,018 | cots<br>1 per person in shelter (planning assumption; no published per-person cot factor cited) | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Floor space (≤72 h) | 80,345 | sq ft<br>Florida DEM (2010): 20 sq ft per person | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Floor space (≤2 weeks) | 160,689 | sq ft<br>Florida DEM (2010): 40 sq ft per person | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Water | 4,018 | gal/day<br>Florida DEM (2010): 1 gal per person per day | American Red Cross | Hawai‘i Department of Human Services (DHS), County Emergency Management, Hawai‘i Department of Education (DOE) – schools, The Salvation Army, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Faith-based organizations, Business community | — | confirm with partner |
| Meals | 8,035 | meals/day<br>Florida DEM (2010): 2 meals per person per day | American Red Cross | Hawai‘i Department of Human Services (DHS), County Emergency Management, Hawai‘i Department of Education (DOE) – schools, The Salvation Army, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Faith-based organizations, Business community | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Toilets (≤72 h) | 101 | toilets<br>Florida DEM (2010): 1 per 40 persons | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Toilets (≤2 weeks) | 201 | toilets<br>Florida DEM (2010): 1 per 20 persons | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Shelter staff | 271 | staff per shift<br>Florida DEM (2010) staffing ratios per shift: registration 1:165, dormitory 1:100, meals 1:110, mental health 1:250, health 1:200, security 1:100, sanitation 1:200, each at least 1; children's area 1:12 | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Office of Hawaiian Affairs (OHA), The Salvation Army, Philanthropy (foundations and donors), Business community | — | confirm with partner |
| Power outlets for power-dependent residents | 26 | outlets<br>1 per power-dependent person in shelter (planning assumption; HHS emPOWER-based estimate) | Hawai‘i Department of Health (DOH) | County Emergency Management, Health care organizations (incl. managed health care), Utilities | — | confirm with partner |
| Accessible transport | 10,265 | people in no-vehicle households at risk<br>ACS via CDC SVI; outreach upper bound, not all will need a ride | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Transportation (DOT), Business community, Other state and county agencies | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Pet space – dog households | 659 | households with a dog AVMA (2020 data): 44.6% of households | County Emergency Management | Hawai‘i VOAD (Voluntary Organizations Active in Disaster) | — | confirm with partner |
| Pet space – cat households | 385 | households with a cat AVMA (2020 data): 26% of households | County Emergency Management | Hawai‘i VOAD (Voluntary Organizations Active in Disaster) | — | confirm with partner |
| Children's area | 809 | children in shelter tract share age 17 and under; staff counted in Shelter staff | County Emergency Management | Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies, Hawai‘i Department of Health (DOH), Health care organizations (incl. managed health care), Philanthropy (foundations and donors) | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Interpreters – Ilocano | 32 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Honolulu); interpreter ratio not sourced – partner to set | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Tagalog/Filipino | 31 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Honolulu); interpreter ratio not sourced – partner to set | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Chinese | 30 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Honolulu); interpreter ratio not sourced – partner to set | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Interpreters – Japanese | 22 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Honolulu); interpreter ratio not sourced – partner to set | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Korean | 12 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Honolulu); interpreter ratio not sourced – partner to set | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Vietnamese | 8 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Honolulu); interpreter ratio not sourced – partner to set | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |

| Need | How many | Unit and basis (source) | Proposed lead | Proposed support | Pledged by partner | Still needed |
|---|---|---|---|---|---|---|
| Interpreters – Chuukese | 7 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Honolulu); interpreter ratio not sourced – partner to set | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Hawaiian | 2 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Honolulu); interpreter ratio not sourced – partner to set | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |
| Interpreters – Marshallese | 1 | people with limited English in shelter CDC SVI limited-English share x ACS PUMS 2019-2023 language mix (Honolulu); interpreter ratio not sourced – partner to set | Hawai‘i Department of Human Services (DHS) | County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies | — | confirm with partner |

Source: likely estimate × planning factors (section 8; Florida Division of Emergency Management 2010; AVMA (2020 data) for pets).

**Service animals stay with their handler (ADA); not counted as pets.**

## Also plan for (county counts, not included in the totals above)

| Group | People | Detail | Date |
|---|---|---|---|
| People experiencing homelessness (Point-in-Time count) | 4,539 | Oʻahu, sheltered and unsheltered, one-night count | 2026-01-26 |
| Visitors on island (average per day) | 107,479 | Oʻahu | 2025 (annual average) |
| People in dam evacuation zones | 37,250 | State dam evacuation zones | zones fetched 2026-10-28 |

Source: Point-in-Time counts – Bridging the Gap 2026 report (Hawaiʻi, Maui, Kauaʻi; night of Jan 25, 2026) and Partners In Care 2026 count for Oʻahu as released by the City and County of Honolulu; visitors – DBEDT State of Hawaiʻi Data Book 2025, Table 7.06. Details: section 8.

State dam evacuation zones show the area that could flood if a dam fails; they are planning zones, not a forecast.

Source: State of Hawaiʻi dam evacuation zones (published through Hawaiʻi Statewide GIS). People = share of each Census area inside a zone × its population, the same method as for the tsunami zone. Shares of people are 2020 Census block populations whose block internal point is inside (Census TIGER internal points), divided by the Census area's 2020 block population; the share is then applied to the area's population.

**Who else to plan for**

- **Domestic-violence survivors:** Served through confidential providers; no counts by design.
- **Off-grid households:** No public Census-area data; county and health partners to supply.
- **People on dialysis or receiving home care:** No public Census-area data; county and health partners to supply.
- **Earthquake:** Not modeled; listed as a limitation in the plan.
- **Caregivers and personal care attendants:** Stay with the person they support; count them for cots, meals and space.
- **Behavioral health:** Mental-health staff at each shelter (ratio: section 8).
- **Niʻihau:** Privately owned island, no public shelter; plan with the landowner and Kauaʻi County.
- **Families afraid to seek shelter (including immigrant families):** The county publishes who may use shelters; trusted community messengers share it.

## 6. Decision points

Proposed – supports, does not replace, county plans.

| When | Trigger | Action | Owner (proposed) | Support |
|---|---|---|---|---|
| **BEFORE** | Annual pre-positioning review (no official product) County (sets the date) | PRE-POSITION: confirm shelters/refuge outside the tsunami evacuation zone (and extreme zone), transport plan for no-vehicle residents inside the zone, pre-approved multilingual evacuation messages. | County Emergency Management (each county) | Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Human Services (DHS), Hawai‘i Department of Transportation (DOT) |
| **BEFORE** | Before each season: who may use shelters (no official product) County (sets the date) | County confirms and publishes who may use shelters and what ID is needed; trusted community messengers share it. | County Emergency Management (each county) | Hawai‘i Department of Human Services (DHS), Faith-based organizations, Office of Hawaiian Affairs (OHA), Hawai‘i VOAD (Voluntary Organizations Active in Disaster) |
| **DURING** | Tsunami Warning NOAA Pacific Tsunami Warning Center (PTWC) | Evacuate the zone; open refuge outside the zone; prioritize residents who cannot evacuate on their own. | County Emergency Management (each county) | Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Transportation (DOT), American Red Cross |
| **AFTER** | Warning cancelled or all clear County / NWS / PTWC | Shelter displaced households, feeding, power for medical devices, donations and volunteers. | American Red Cross | Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Hawai‘i Department of Health (DOH), Utilities, The Salvation Army |

Source: National Weather Service, Central Pacific Hurricane Center and Pacific Tsunami Warning Center product definitions (links: section 8).

# 7. Priority outreach and languages

## Top 10 Census areas by number of people with needs

| # | Community (Census area) | People in the risk area | Use power-dependent medical equipment (estimate) | People with a disability | Speak English less than well | People in households without a car | Needs total (a person can be in more than one column) |
|---|---|---|---|---|---|---|---|
| 1 | Waiʻanae Census area 97.07 | 3,933 ± 639 | 18 | 700 ± 178 | 40 ± 53 | 1,109 ± 384 | 1,868 |
| 2 | Urban Honolulu Census area 37.02 | 3,136 ± 632 | 26 | 587 ± 378 | 303 ± 143 | 522 ± 413 | 1,438 |
| 3 | Urban Honolulu Census area 19.06 | 3,078 ± 392 | 24 | 341 ± 113 | 259 ± 112 | 640 ± 211 | 1,263 |
| 4 | Urban Honolulu Census area 37.01 | 2,509 ± 376 | 15 | 306 ± 165 | 399 ± 224 | 523 ± 248 | 1,243 |
| 5 | Urban Honolulu Census area 19.07 | 2,604 ± 709 | 22 | 292 ± 110 | 268 ± 200 | 585 ± 187 | 1,166 |
| 6 | Mākaha Census area 98.03 | 2,014 ± 548 | fewer than 10 | 306 ± 98 | 419 ± 422 | 306 ± 101 | 1,039 |
| 7 | Urban Honolulu Census area 18.03 | 1,597 ± 199 | 11 | 168 ± 69 | 80 ± 49 | 707 ± 166 | 966 |
| 8 | Māʻili Census area 96.10 | 6,167 ± 1,333 | 33 | 765 ± 333 | 16 ± 88 | 103 ± 166 | 918 |
| 9 | Lāʻie Census area 102.02 | 5,026 ± 783 | 14 | 328 ± 92 | 59 ± 60 | 508 ± 325 | 910 |

| # | Community (Census area) | People in the risk area | Use power-dependent medical equipment (estimate) | People with a disability | Speak English less than well | People in households without a car | Needs total (a person can be in more than one column) |
|---|---|---|---|---|---|---|---|
| 10 | Nānākuli<br>Census area 96.08 | 3,053 ± 422 | 13 | 397 ± 125 | fewer than 10 | 367 ± 219 | 786 |

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

Per-area figures – People in the risk area: ACS 2018–2022, table S0601 (via CDC/ATSDR SVI 2022); Use power-dependent medical equipment (estimate): HHS emPOWER county count, allocated by each area's share of people age 65 and older (estimate; no margin of error); People with a disability: ACS 2018–2022, table DP02 (DP02_0072E, via CDC/ATSDR SVI 2022); Speak English less than well: ACS 2018–2022, table B16005 (via CDC/ATSDR SVI 2022); People in households without a car: ACS 2018–2022, table DP04 (DP04_0058E, households without a vehicle, via CDC/ATSDR SVI 2022) × people per household; margin of error scaled the same way. ± is the 90% margin of error published with the estimate (SVI M_ columns; ACS), scaled by the fraction of residents who live in the risk area. Shelter estimates and need totals are model results with no margin of error. Per-area counts from 1 to 9 are not shown. All figures are public Census Bureau, CDC and HHS estimates, not personal records.

## More needs in the largest communities

| Community (Census area) | Hearing difficulty | Vision difficulty | Cognitive difficulty (memory, decisions) | Age 65 and older, living alone |
|---|---|---|---|---|
| Māʻili<br>Census area 96.10 | 411 ± 223 | 91 ± 72 | 403 ± 157 | 89 ± 54 |
| Lāʻie<br>Census area 102.02 | 44 ± 40 | 79 ± 57 | 186 ± 67 | fewer than 10 |
| Waiʻanae<br>Census area 97.07 | 252 ± 103 | 101 ± 74 | 321 ± 101 | 74 ± 42 |

Source: ACS 2020–2024 5-year Census-area tables B18102 (hearing), B18103 (vision), B18104 (cognitive) and B09020 (age 65 and older living alone), times the share of each area in the risk area; homes in 10+ unit buildings: CDC/ATSDR SVI 2022 (E_MUNIT, housing units).

Per-area figures – Hearing difficulty: ACS 2020–2024, table B18102; Vision difficulty: ACS 2020–2024, table B18103; Cognitive difficulty (memory, decisions): ACS 2020–2024, table B18104; Age 65 and older, living alone: ACS 2020–2024, table B09020. ± is the 90% margin of error published with the estimate (SVI M_ columns; ACS), scaled by the fraction of residents who live in the risk area. Shelter estimates and need totals are model results with no margin of error. Per-area counts from 1 to 9 are not shown. All figures are public Census Bureau, CDC and HHS estimates, not personal records.

## How to reach residents

| Channel | Owner (proposed) | Support (proposed) |
|---|---|---|
| Official alerts: Wireless Emergency Alerts / Emergency Alert System and county alert sign-ups | County Emergency Management (each county) | Hawaiʻi Emergency Management Agency (HI-EMA) |

| Channel | Owner (proposed) | Support (proposed) |
|---|---|---|
| Broadcast radio, including in-language programs where available | County Emergency Management (each county) | Hawai‘i Emergency Management Agency (HI-EMA) |
| Outdoor warning sirens: a signal to turn on radio, TV or phone alerts for instructions | Hawai‘i Emergency Management Agency (HI-EMA) | County Emergency Management |
| Trusted community messengers: faith-based groups, Native Hawaiian organizations, health care providers, community nonprofits | Hawai‘i Department of Human Services (DHS) | Faith-based organizations, Office of Hawaiian Affairs (OHA), Health care organizations (incl. managed health care), Hawai‘i VOAD (Voluntary Organizations Active in Disaster) |
| Printed notices at shelters, libraries and schools, in the county's top languages | County Emergency Management (each county) | Hawai‘i State Public Library System, Hawai‘i Department of Education (DOE) – schools |

Source: proposed channels and formats in the plan's communication pathways – supports, does not replace, county and State public-information plans.

Populated Census areas whose land point is farther than 3,400 ft from every outdoor siren (HI-EMA manufacturer rating; actual range varies): **110**.

**Populated Census areas with less than 50% of people within rated siren range (74): prioritise WEA, radio and trusted messengers.** East Honolulu (1.15, 1.16, 1.18, 1.21, 2, 4.01); Helemano / Haleʻiwa (100); Hickam Housing (71, 73.02, 9819); Kahaluʻu (103.03); Kailua (111.04); Kalaeloa (85.02); Kaneohe Base (9817, 9818.01, 9818.02, 9818.03); Ko Olina (86.10); Kāneʻohe (105.08, 105.09); Makakilo (86.30, 86.31, 86.34); Mililani Mauka (89.31); Mākaha (98.01); Nānākuli (9400.05, 9400.07); Pearl City (78.09, 78.14, 78.15, 114.01); Royal Kunia (89.34, 89.50); Schofield Barracks (95.07, 95.08, 95.09, 95.10, 95.11, 95.12); Urban Honolulu (10, 12.01, 22.06, 23.01, 27.01, 28.01, 29, 46, 65, 67.04, 68.11, 68.12, 68.13, 68.14, 68.15, 68.18, 68.19, 69, 70.01, 70.02, 9802, 9813, 9820); Wahiawā (94.01); Waipahu (87.05, 88.02); Waipiʻo Acres (89.39); Waiʻanae (97.06); Wheeler AFB (90); near Hālawa (74); near Kailua (9811); near Ko Olina (86.11); near Pearl City (9822); near Schofield Barracks (9806); rural near Waiʻanae (97.04)

Source: Hawai‘i Emergency Management Agency public siren map (locations of permanent outdoor warning sirens; no live status). Rated range: 3,400 feet, the manufacturer rating published by the Hawai‘i Emergency Management Agency; the real range varies with terrain, buildings and weather. In range: within 3,400 feet of a siren, the same method as for the tsunami zone. Shares of people are 2020 Census block populations whose block internal point is inside (Census TIGER internal points), divided by the Census area's 2020 block population; the share is then applied to the area's population.

### Accessible formats for every message:

- **ASL video:** short American Sign Language video of each message
- **Captions:** captions on every video and broadcast
- **Large print:** printed notices in large print
- **Plain language:** short sentences, one action at a time
- **Screen-reader-friendly text:** plain text or tagged web pages, not images of text

## Languages: people age 5 and older who speak English less than "very well" (Honolulu)

| Language | People | Margin of error (90%) |
|---|---|---|
| Ilocano | 21,729 | ±1,818 |
| Tagalog/Filipino | 21,584 | ±1,870 |
| Chinese | 20,695 | ±1,997 |
| Japanese | 14,936 | ±1,459 |
| Korean | 8,555 | ±1,020 |
| Vietnamese | 5,648 | ±1,522 |
| Chuukese | 4,948 | ±1,848 |
| Hawaiian | 1,318 | ±348 |
| Marshallese | 1,021 | ±345 |

Source: ACS PUMS 2019–2023 5-year (validated against the DBEDT State of Hawai‘i Data Book 2024, Table 1.44).

Draft messages are available statewide in Chinese, Hawaiian, Ilocano, Japanese, Korean, Pidgin, Spanish, Tagalog and Vietnamese; Chuukese and Marshallese need translators.

## All languages

All languages in the survey for Honolulu: people age 5 and older who speak English less than "very well"

| Language | People | Margin of error (90%) | Reliability |
|---|---|---|---|
| Ilocano | 21,729 | ±1,818 | |
| Tagalog/Filipino | 21,584 | ±1,870 | |
| Chinese | 20,695 | ±1,997 | |
| Japanese | 14,936 | ±1,459 | |
| Korean | 8,555 | ±1,020 | |
| Vietnamese | 5,648 | ±1,522 | |
| Chuukese | 4,948 | ±1,848 | |
| Spanish | 3,543 | ±610 | |
| Samoan | 2,721 | ±716 | |
| Other Eastern Malayo-Polynesian languages | 1,810 | ±728 | |
| Cebuano and other Philippine languages | 1,578 | ±481 | |
| Hawaiian | 1,318 | ±348 | |
| Thai | 1,036 | ±353 | |
| Marshallese | 1,021 | ±345 | |
| Tongan | 613 | ±257 | |
| French | 587 | ±218 | |
| Lao | 581 | ±247 | |
| German | 460 | ±232 | |
| Pidgin (Hawai‘i Creole English) | 460 | ±240 | |
| Russian | 348 | ±193 | |
| Khmer | too few to estimate | | low reliability |

| Language | People | Margin of error (90%) | Reliability |
|---|---|---|---|
| Portuguese | 180 | ±112 | |
| Swahili | too few to estimate | | low reliability |
| Telugu | too few to estimate | | low reliability |
| Indonesian | 156 | ±90 | |
| Tamil | too few to estimate | | low reliability |
| Punjabi | too few to estimate | | low reliability |
| Hindi | too few to estimate | | low reliability |
| Nepali | too few to estimate | | low reliability |
| Burmese | too few to estimate | | low reliability |
| India N.E.C. | too few to estimate | | low reliability |
| Romanian | too few to estimate | | low reliability |
| Other Native North American languages | too few to estimate | | low reliability |
| Chamorro | too few to estimate | | low reliability |
| Polish | too few to estimate | | low reliability |
| Arabic | too few to estimate | | low reliability |
| Serbian | too few to estimate | | low reliability |
| Bengali | too few to estimate | | low reliability |
| Hmong | too few to estimate | | low reliability |
| Mongolian | too few to estimate | | low reliability |
| Swedish | too few to estimate | | low reliability |
| Italian | too few to estimate | | low reliability |
| Other and unspecified languages | too few to estimate | | low reliability |
| Armenian | too few to estimate | | low reliability |
| Amharic | too few to estimate | | low reliability |
| Sinhala | too few to estimate | | low reliability |
| Bulgarian | too few to estimate | | low reliability |
| Tigrinya | too few to estimate | | low reliability |
| Tibetan | too few to estimate | | low reliability |
| Hungarian | too few to estimate | | low reliability |
| Albanian | too few to estimate | | low reliability |
| Haitian | too few to estimate | | low reliability |
| Other Niger-Congo languages | too few to estimate | | low reliability |
| Gujarati | too few to estimate | | low reliability |
| Other Bantu languages | too few to estimate | | low reliability |
| Navajo | too few to estimate | | low reliability |
| Ukrainian | too few to estimate | | low reliability |
| Chin languages | too few to estimate | | low reliability |
| Afrikaans | too few to estimate | | low reliability |
| Akan (including Twi) | too few to estimate | | low reliability |
| Assyrian Neo-Aramaic | too few to estimate | | low reliability |
| Bosnian | too few to estimate | | low reliability |
| Cherokee | too few to estimate | | low reliability |
| Croatian | too few to estimate | | low reliability |
| Czech | too few to estimate | | low reliability |
| Dakota languages | too few to estimate | | low reliability |

| Language | People | Margin of error (90%) | Reliability |
|---|---|---|---|
| Danish | too few to estimate | | low reliability |
| Dutch | too few to estimate | | low reliability |
| Farsi | too few to estimate | | low reliability |
| Ga | too few to estimate | | low reliability |
| Hebrew | too few to estimate | | low reliability |
| Iu Mien | too few to estimate | | low reliability |
| Jamaican Creole English | too few to estimate | | low reliability |
| Kannada | too few to estimate | | low reliability |
| Lithuanian | too few to estimate | | low reliability |
| Macedonian | too few to estimate | | low reliability |
| Malay | too few to estimate | | low reliability |
| Malayalam | too few to estimate | | low reliability |
| Manding languages | too few to estimate | | low reliability |
| Norwegian | too few to estimate | | low reliability |
| Other Indo-European languages | too few to estimate | | low reliability |
| Other Indo-Iranian languages | too few to estimate | | low reliability |
| Other languages of Asia | too few to estimate | | low reliability |
| Serbocroatian | too few to estimate | | low reliability |
| Shona | too few to estimate | | low reliability |
| Swiss German | too few to estimate | | low reliability |
| Turkish | too few to estimate | | low reliability |
| Urdu | too few to estimate | | low reliability |
| Uto-Aztecan languages | too few to estimate | | low reliability |
| Yoruba | too few to estimate | | low reliability |

Source: ACS PUMS 2019–2023 5-year (validated against the DBEDT State of Hawai‘i Data Book 2024, Table 1.44).

Low reliability: margin of error large compared with the estimate (coefficient of variation above 40%), or too few to estimate (fewer than 100 people or 10 survey responses).

Pre-approved message templates: none yet; no machine translation without human review.

# 8. Sources and assumptions

## Planning factors

| Factor | Value | Source or team assumption |
|---|---|---|
| Shelter-seeking rate – Hurricane | 2% / 3% / 5% | South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%) |
| Shelter-seeking rate – Wildfire | 2% / 5% / 11.2% | high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are Team assumption. |
| Shelter-seeking rate – Tsunami | 2% / 5% / 11% | Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Shelter-seeking rate – Inland flooding | 2% / 5% / 11% | Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Shelter-seeking rate – Coastal flooding | 2% / 5% / 11% | Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Footprint – Hurricane | Every Census area | Hurricanes affect whole islands (Lala, Lowell 2026). |
| Footprint – Tsunami | Share living inside the tsunami evacuation zone | Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately. |
| Footprint – Wildfire | Top 25% of Census areas by FEMA risk score | Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption). |
| Footprint – Inland flooding | Top 25% of Census areas by FEMA risk score | Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption). |
| Footprint – Coastal flooding | Top 25% of Census areas by FEMA risk score | Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption). |
| Floor sqft 72h | 20 | Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h |
| Floor sqft 2wk | 40 | Florida DEM (2010): short-term shelter, <=2 weeks |
| Water gal per day | 1 | Florida DEM (2010) |
| Meals per day | 2 | Florida DEM (2010): two hot meals a day plus snacks |

| Factor | Value | Source or team assumption |
|---|---|---|
| Persons per toilet short term | 20 | Florida DEM (2010): short-term shelter, <=2 weeks |
| Power outlets per power dependent person | 1 | Team assumption |
| Persons per toilet 72h | 40 | Florida DEM (2010): emergency evacuation shelter, <=72 h |
| Staffing ratios (one staff member per N residents) | client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12 | One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations. |
| Dog owning household share | 0.446 | AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27. |
| Cat owning household share | 0.26 | AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'. |

## Data sources

| Source | Fetched | URL / file |
|---|---|---|
| CDC/ATSDR Social Vulnerability Index 2022, census tracts | 2026-10-27 | https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2 |
| FEMA National Risk Index v1.20, census tracts – Hurricane | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Index_Rating/FeatureServer/0 |
| FEMA National Risk Index v1.20, census tracts – Wildfire | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index_Rating/FeatureServer/0 |
| FEMA National Risk Index v1.20, census tracts – Tsunami | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Index_Rating/FeatureServer/0 |
| FEMA National Risk Index v1.20, census tracts – Inland Flooding | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Risk_Index_Rating/FeatureServer/0 |

| Source | Fetched | URL / file |
|---|---|---|
| FEMA National Risk Index v1.20, census tracts – Coastal Flooding | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_Risk_Index_Rating/FeatureServer/0 |
| Tsunami Evacuation Zones (Statewide GIS) | 2026-10-27 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2 |
| Extreme Tsunami Evacuation Zones (Statewide GIS) | 2026-10-27 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12 |
| HHS emPOWER – electricity-dependent DME, county level | 2026-10-27 | https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2 |
| ACS 2020-2024 5-year, tract tables B18102 (hearing), B18103 (vision), B18104 (cognitive), B09020 (age 65+ living alone) | 2026-10-26 | https://data.census.gov/api/access/data/table?g=040XX00US15%241400000&id=ACSDT5Y2024.B18102 |
| Share of people in a zone (tsunami, dam evacuation zones, siren range) | | Shares of people are 2020 Census block populations whose block internal point is inside (Census TIGER internal points), divided by the Census area's 2020 block population; the share is then applied to the area's population. |
| Hawai‘i Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26) | 2026-10-28 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6 |
| County shelter activations 2026 (official county releases) | 2026-10-27 | County shelter activation table: one official county source per row |
| 2020 Census Block Assignment File, place/CDP (tract community labels) | 2026-10-27 | https://www2.census.gov/geo/docs/maps-data/data/baf2020/BlockAssign_ST15_HI.zip |
| 2020 Census PL 94-171 block population P1_001N (tract community labels) | 2026-10-27 | https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/Hawaii/hi2020.pl.zip |
| ACS PUMS 2019-2023 5-year (detailed languages) | 2026-10-27 | https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip |
| DBEDT State of Hawai‘i Data Book 2024, Table 1.44 (validation) | 2026-10-27 | https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf |
| Partner list: Hawai‘i DHS – ESF-6 challenge statement (2026) | 2026-10-15 | Hawai‘i DHS – ESF-6 challenge statement (2026) |
| pre season hurricane | | https://www.nhc.noaa.gov/cphcmission.php |
| hurricane watch | | https://www.nhc.noaa.gov/aboutgloss.shtml |
| hurricane warning | | https://www.nhc.noaa.gov/aboutgloss.shtml |
| tsunami warning | | https://www.tsunami.gov/?page=message_definitions |
| flash flood warning | | https://www.weather.gov/safety/flood-watch-warning |
| red flag warning | | https://www.weather.gov/safety/wildfire-ww |

| Source | Fetched | URL / file |
|---|---|---|
| coastal flood warning | | https://www.weather.gov/okx/marinedef |
| coastal flood warning in effect | | https://www.weather.gov/okx/marinedef |
| Ready Keiki Locator – All Pre-K Sites (Office of the Lieutenant Governor, hosted by Hawai‘i Statewide GIS) | 2026-10-28 | https://services.arcgis.com/HQ0xoN0EzDPBOEci/arcgis/rest/services/Ready_Keiki_GIS/FeatureServer/0 |
| Hawai‘i Emergency Management Agency – all-hazard outdoor warning sirens (public siren layer; locations only) | 2026-10-28 | https://services9.arcgis.com/aKxrz4vDVjfUwBWJ/arcgis/rest/services/public_siren_status_layer_view/FeatureServer/0 |
| State of Hawai‘i dam evacuation zones (hosted by Hawai‘i Statewide GIS) | 2026-10-28 | https://services.arcgis.com/HQ0xoN0EzDPBOEci/arcgis/rest/services/DamEvacZones_joined/FeatureServer/0 |
| Hawai‘i Emergency Management Agency – siren audible radius (3,400 feet, manufacturer rating) | 2026-10-28 | https://dod.hawaii.gov/hiema/all-hazard-statewide-outdoor-warning-siren-system/ |
| Tsunami Safe Zones (Hawai‘i Statewide GIS, Hazards layer 13) (Hawai‘i Island March 2025 (corrected November 2025), O‘ahu April 2025, Kaua‘i March 2025, Maui, Moloka‘i and Lāna‘i 2013) | 2026-10-30 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/13 |
| Fire Risk Areas (DLNR Division of Forestry and Wildlife, via Hawai‘i Statewide GIS, Hazards layer 7) (2007; boundaries confirmed by DLNR-DOFAW 2017 and 2022) | 2026-10-30 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/7 |
| Sea Level Rise – Potential Flooded Highways, 3.2 ft scenario (Hawai‘i Sea Level Rise Viewer, PacIOOS, via Hawai‘i Statewide GIS, Climate layer 71) (Hawai‘i Sea Level Rise Vulnerability and Adaptation Report, 2017) | 2026-10-30 | https://geodata.hawaii.gov/arcgis/rest/services/Climate/MapServer/71 |

| Source | Fetched | URL / file |
|---|---|---|
| Pacific Disaster Center – Hawaii Emergency Shelters (capacity: American Red Cross Hawai‘i State Chapter) (ground condition 2009-10-08 (PDC metadata)) – via third-party ArcGIS Online copy; PDC catalogue confirms 2009 date | 2026-10-30 | https://services1.arcgis.com/x4h61KaW16vFs7PM/arcgis/rest/services/Potential_Emergency_Shelters_PDC/FeatureServer/36 |
| Hawai‘i DOT Highway Performance Monitoring System road centrelines (Hawai‘i Statewide GIS, Transportation layer 12) | 2026-10-30 | https://geodata.hawaii.gov/arcgis/rest/services/Transportation/MapServer/12 |
| Hawai‘i DOT surveyed mile markers (road-closure point, Route 11 in Ka‘ū) | 2026-10-30 | https://geodata.hawaii.gov/arcgis/rest/services/Transportation/MapServer/14 |
| DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census) | 2026-10-27 | https://files.hawaii.gov/dbedt/economic/databook/db2025/DB2025_section07.xlsx |
| CMS Provider Data Catalog – Dialysis Facility – Listing by Facility | | https://data.cms.gov/provider-data/dataset/23ew-n7w9 |
| CMS Provider Data Catalog – Nursing Homes – Provider Information | | https://data.cms.gov/provider-data/dataset/4pq5-n9py |
| CMS Provider Data Catalog – Hospital General Information | | https://data.cms.gov/provider-data/dataset/xubh-q36u |
| Esri ArcGIS World Geocoder (site check, care facilities) and ArcGIS routing (drive and walk times, normal traffic). Powered by Esri. | | https://developers.arcgis.com/rest/ |

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only. Per-area counts from 1 to 9 are not shown. All figures are public Census Bureau, CDC and HHS estimates, not personal records. In a Census area with fewer than 100 residents only the population is shown.

## 9. Results

- **Coverage:** 436 of 436 Census areas have needs data (FEMA risk data joined for 435 of 436). Source: coverage check of the needs layer (CDC SVI 2022, FEMA NRI v1.20).
- **Backtest – Lahaina wildfire (2023-08-11):** observed 1,418 people in shelters; model 254 / 635 / 1,423 (low / likely / high). This is a calibration event, not validation: the high rate was set from it. Source: Lahaina 2023 backtest (Maui Now count of people in shelters).

# 10. Stress tests – summary

**Stress test, not a forecast.** Each test changes one thing in the plan (a site closes, two hazards strike together, a road is cut, more people come) and counts who is affected, from the plan's own data. Deterministic code computes every number from the plan's data; people review the results.

People counted: residents of Census areas whose drive from the area's centre point to the nearest usable 2026 site was 15 minutes or less before the loss and is more than 15 minutes (or no route) after it; areas already beyond 15 minutes are not counted.

Top 3 for Honolulu – Tsunami and all-hazard tests, ranked by people affected (at most two from one test). Full results: the **Honolulu stress-test annex** (a separate PDF, also linked in the app).

1. **Extreme tsunami zone (worst case) – Honolulu**
   208,201 people in the extreme tsunami evacuation zone (may need a public shelter (likely): 10,410; 2026 sites inside the extreme zone: 1).
2. **Visitors on the island(s) – Honolulu (tsunami)**
   107,479 visitors on an average day (island level) (visitors inside the tsunami zone: needs county data (hotel and rental locations)).
3. **Stranded visitors for 3 or 7 days – Honolulu**
   107,479 visitors on an average day (island level).

Sources, evidence and team assumptions for each result: in the annex.

## Questions for counties

These need county knowledge. The plan has no data for them, so it shows no numbers.

1. **Shelter buildings in an earthquake or a strong hurricane:** Which shelter buildings have been assessed for earthquake and hurricane-wind resilience, and which need a check before they are used after a strong event?
2. **Resupply for Molokaʻi and Lānaʻi:** How would water, food, cots, fuel and medical supplies reach Molokaʻi and Lānaʻi if the harbours or airports close for several days, and who decides the order of shipments?
3. **Cell network outages:** Which areas lose mobile phone service first in a long power outage or after wind damage, and how will warnings and shelter information reach them (radio, sirens, door-to-door)?
4. **Schools used as shelters:** When a school is used as a shelter, how long can it stay open before classes must restart, and where do residents go next?
5. **Partner capacity:** Could each county collect partner capacity (cots, meals, staff, vehicles, interpreters) with a county-run form such as Survey123, with no personal data, so the plan can show pledged capacity instead of 'confirm with partner'?

## 11. After the event

### Closing shelters and moving people to temporary housing

The county decides when each shelter closes. Before a shelter closes, every resident should have a next place to stay: home, family or friends, or temporary housing arranged with the county, Hawai‘i DHS, the Red Cross and, after a federal declaration, FEMA. People who use power-dependent equipment, people with a disability and people without a car need a confirmed place and a ride before they leave.

### Reconnecting residents to programs

Partners can help residents reconnect to SNAP and Med-QUEST (Hawai‘i DHS) and to disaster assistance (FEMA and the U.S. Small Business Administration after a federal declaration; county and State programs). Program rules can change after a disaster: check current rules with the agency. This playbook makes no eligibility claims.

### Language access in recovery

Recovery notices, forms and help desks need the same languages as the warnings in section 7. Ask for interpreters at recovery centers and for applications, through the Hawai‘i State Office of Language Access and community partners.

### When to rerun the plan

Rerun this plan after shelters close, when the county changes its shelter list or capacity, when partners change roles, and when new Census or CDC data are released.